IRON-MAN SERVO

PRO

BY WALSH3D

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Intro

"Motorization"
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The "Iron-Man Servo Pro" is an all-in-one motorization kit for 3D printed helmets that doesn't rely on the typical "helper arms" setup.

The purpose of this is to reduce the difficulty of assembly, streamline the motorization process, allow for more complex motorization systems, and provide a 100% universal kit for all helmets.

The design of the "Iron-Man Servo Pro" was derived from the average opening angle and arm length of several Iron-Man helmets by different creators. This allowed Walsh3D to determine the best-fit design for all Iron-Man helmets.

This kit also features:
1] optional "HUD" face lighting
2] scalable mounting system
3] installation alignment tool
4] "hotswappable" design with 4 screws
5] quick disassembly knob
6] wiring channel
7] locknuts to reduce screw loosening

Bill of Materials

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Bill of Hardware			
Category	**Part Name**	**Notes**	**Number**
Servo	MG90S digital servo	180°	2
Hardware	M3x8 SHCS		4
Hardware	M3x10 SHCS		2
Hardware	M3x12 SHCS		2
Hardware	M3x16 SHCS		2
Hardware	M3x45 SHCS		1
Hardware	M3 locknuts		4
Hardware	OD 5mm M3 heat-set inserts	"RX-M3x5x4"	11
LED	5mm LED	(For face lighting)	2

Bill of Electronics (consult Crashworks "Iron-Man Servo" GitHub)			
Category	**Part Name**	**Notes**	**Number**
Micro-controller	Arduino Nano		1
Micro-controller	Crashworks 3D "A.L.I.S.H.A."	Alternative to Arduino	
Button	Limit Switch	Or other button alternative	1
Battery	5V Power Bank	Or equivalent arduino battery	1

Print Settings

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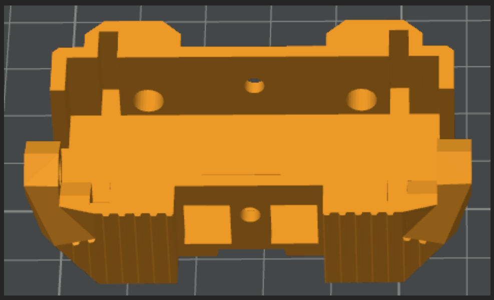
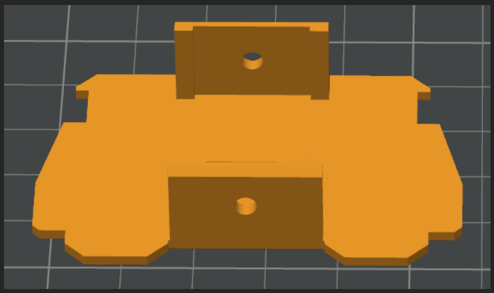
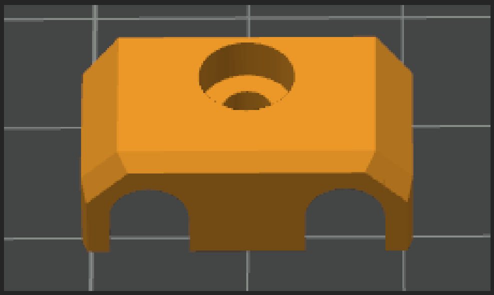
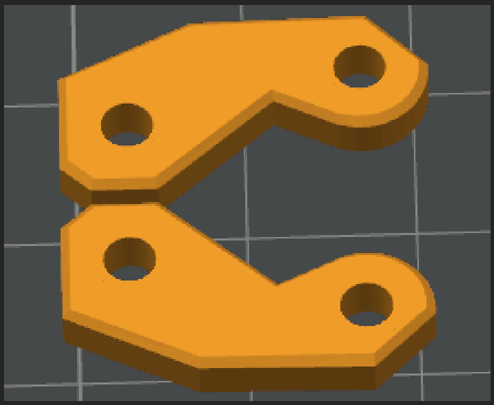
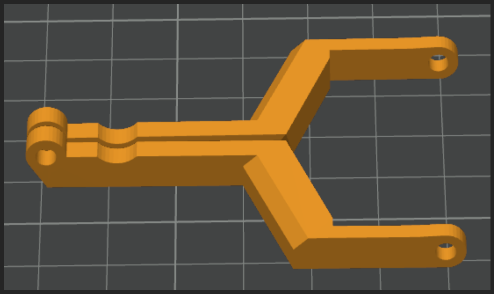
This section will cover the print settings and orientation for each of the motorization kit components.

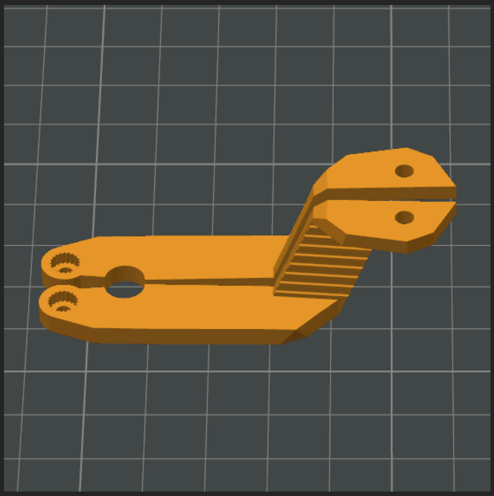
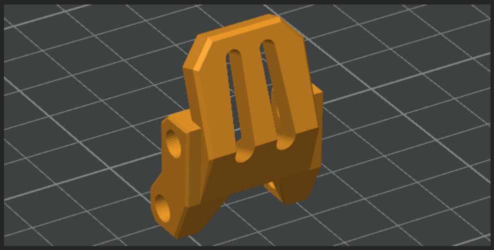
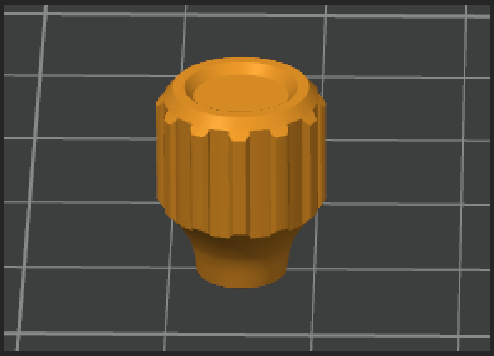
Print Settings	
Layer Height	0.16mm
Infill	40%
Wall Count	4
Support Material	When instructed

Keep all of the following parts at 100% scale:

Part Name	**Notes**
Arm Retainer	Mirror and print 2
Face Lighting Module	
Faceplate Interface	
Knob	
Primary Servo Arm	Mirror and print 2
Secondary Servo Arm	Mirror and print 2
Servo Case	
Servo Cover	

Orientation

Part	Notes	Supports	Picture
Servo Case	Flat on bottom surface	No	
Servo Cover	Flat on lid with sides sticking up	No	
Face Lighting Module	Flat on back side (upside down is also acceptable)	No	
Arm Retainer	Print on flat side Mirror one and print 2	No	
Secondary Servo Arm	Print on flat side Mirror one and print 2	No	

Primary Servo Arm	Print with long side facing down and servo gear facing up <u>Mirror one and print 2</u> Do not worry if the gear teeth don’t print, it will still work	Yes	
Faceplate Interface	Print as vertical as possible	Yes	
Knob	Print on bottom with insert hole down.	No	

Scaling

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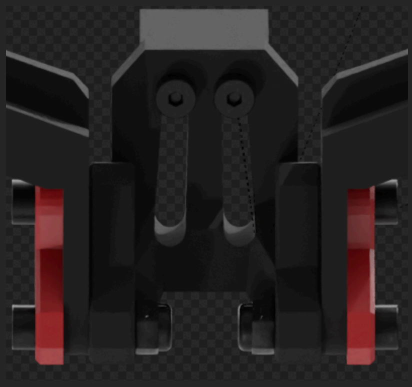
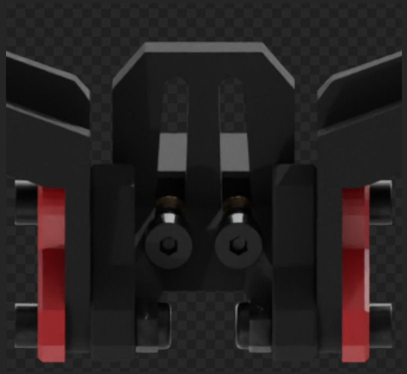
The "Iron-Man Servo Pro" contains a unique scaling system to make sure everything lines up perfectly.

Each of the mounting holes are a cam shape so that as the helmet is scaled with the mount attached, the screw holes still line up and are the proper size.

This works between 90% and 110% scale from default.

Mount Scale Example (reference "scaling" animation in renders folder)		
Mount (with helmet) at 90% scale		
Mount (with helmet) at 110% scale		

Assembly

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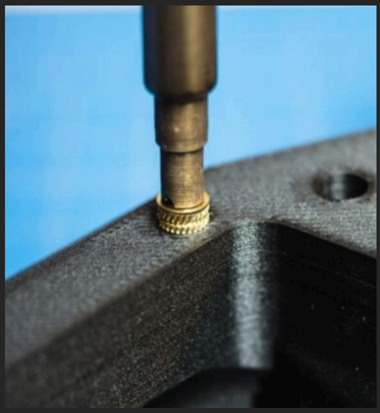
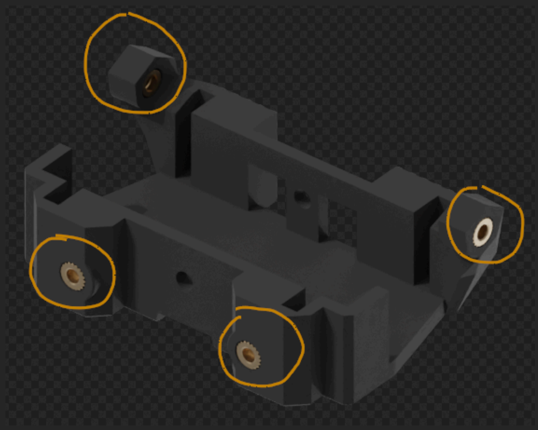
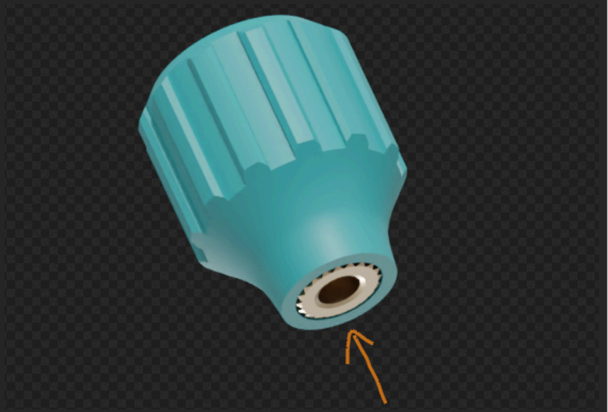
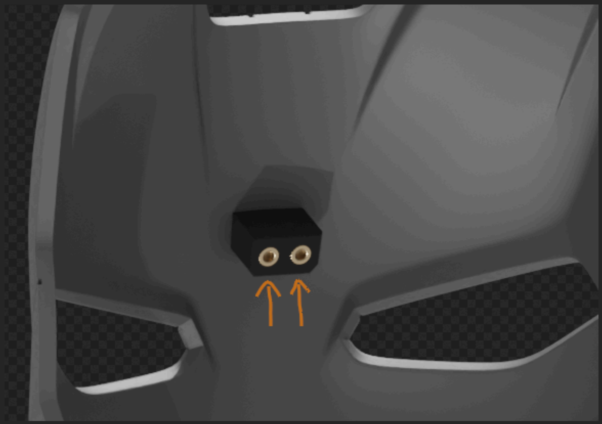
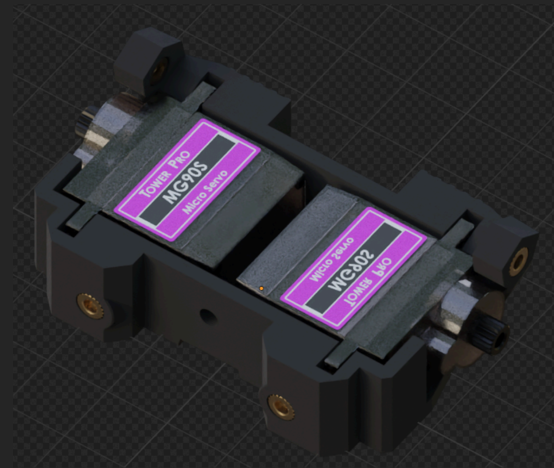
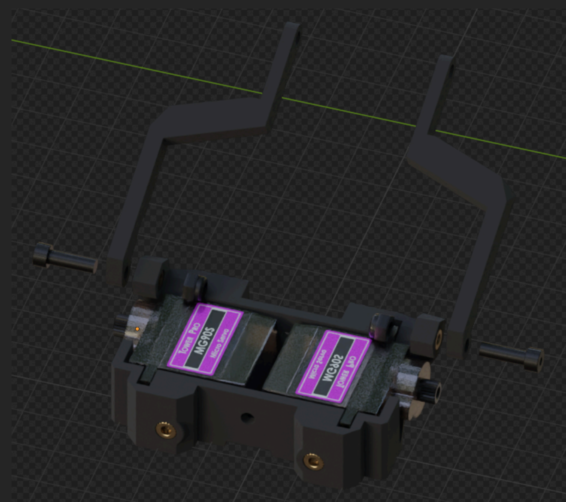
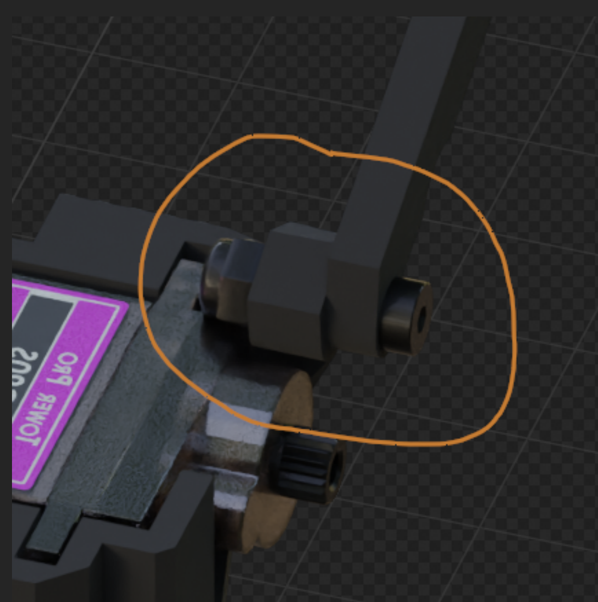
This section will cover the assembly of the "Iron-Man Servo Pro."

Before assembly though, we need to establish a key concept:

Inserting M3 Heat-set Inserts		
Concept	**Description**	**Picture**
M3 Heat-set inserts Insertion	In order to insert M3 heat-set inserts, you align the small side of the insert with the provided hole, and gently push it in with a soldering iron. <u>Do not push the insert in too far.</u>	

Assembling the Kit

Step	Description	Pictures
1]	Insert 4 M3 heat set inserts into the “Servo Case”	
2]	Insert 4 M3 heat set inserts into the “Servo Interface”	
3]	Insert 1 heat set insert into the “knob”	
4]	Insert 2 heat set insert into the “faceplate mount” on the faceplate	

5]	Place 2 MG90S servos into the "Servo Case"	
6]	Place the "Secondary Servo Arms" onto the mount and screw them in using an M3x12 screw and M3 locknut on the inside	
7]	The assembly should look like this	

8]	Place the servo cover onto the servo case assembly	
9]	Place two LEDs into the "Face Lighting Module"	
10]	Insert an M3x45 screw through the servo case assembly	

11]	Screw the knob on the other side of the M3x45 to complete the servo case assembly	
12]	This step you'll need to line up several parts at once. Place the "Secondary Servo Arm" and the "Primary Servo Arm" onto the "faceplate interface" The "arm retainer" will go overtop of it and screw in using an M3x12 for the "Primary Servo Arm" and an M3x16 for the "Secondary Servo Arm" (keep in mind the secondary arms are attached to the servo case)	
13]	The completed assembly should look like this. (keep in mind the secondary arms are attached to the servo case)	

14]	On the backside, screw an M3 locknut onto the other end of the “Secondary Servo Arm” screw	
15]	Test the motion with your hands, if any of the arms are too tight to move freely, loosen the screw by half a turn	
16]	Before attaching the “Primary Servo Arm,” the servo must be zero’d. Below will explain the process of zero’ing your Servos.	

Zero’ing Your MG90S

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Without running any code, an MG90S doesn’t “know” what position it is at– moving the servo by hand to the furthest counterclockwise position is not suitable.

The purpose of this procedure is to bring your servo to the “closed” position before attaching the servo arms.

To do this, just <u>run your code once and turn it off in the closed position</u>.

Alternatively, if you plan to do your own code follow the instructions below:

Servo1 will move between 0° and 170° (0° being the closed position)
Servo2 will move between 180° and 10° (180° being the closed position)

If you are using these angles, you will want to Zero them at 10° and 170° respectively. This is to make sure the servo mount fully seats itself in the “closed” position.

Zero’ing Code

```
#include <Servo.h>

void setup() {
  // put your setup code here, to run once:
Servo servo1;
Servo servo2
servo1.attach(9); //attach Servo 1 to Pin 9 on Arduino
servo2.attach(10); //attach Servo 2 to Pin 10 on Arduino
}

void loop() {
  // put your main code here, to run repeatedly:
servo1.write(10); //move servo1 to 10 degrees
servo2.write(170); //move servo2 to 170 degrees
}
```

Assembly cont.

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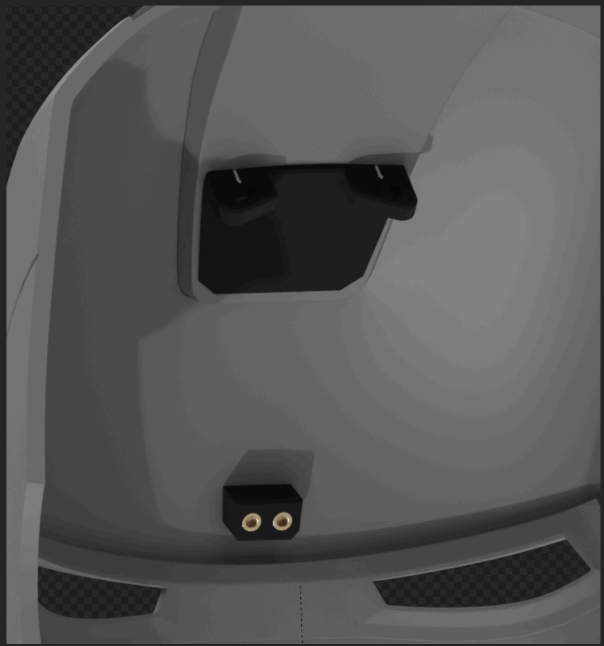
Assembling the Kit cont.		
Step	**Description**	**Pictures**
17]	Push the "Primary Servo Arms" onto the output spline of the MG90S servos You may have to push them hard against a flat surface– that is okay The teeth on the servo will self-tap into the plastic Screw them in with the small screw that came with your servos	
18]	Now we will mount the kit using the mounting system in your helmet If you are using a helmet without the mount pre-installed, there are instructions in this documentation for adding the mounting system	

19]	On the dome of the helmet, screw in the kit using 2 M3x8 screws	
20]	On the faceplate of the helmet, screw in the kit using 2 M3x8 screws	
Done!	Now you're all finished with the assembly of the "Iron-Man Servo Pro"	

Code / Wiring

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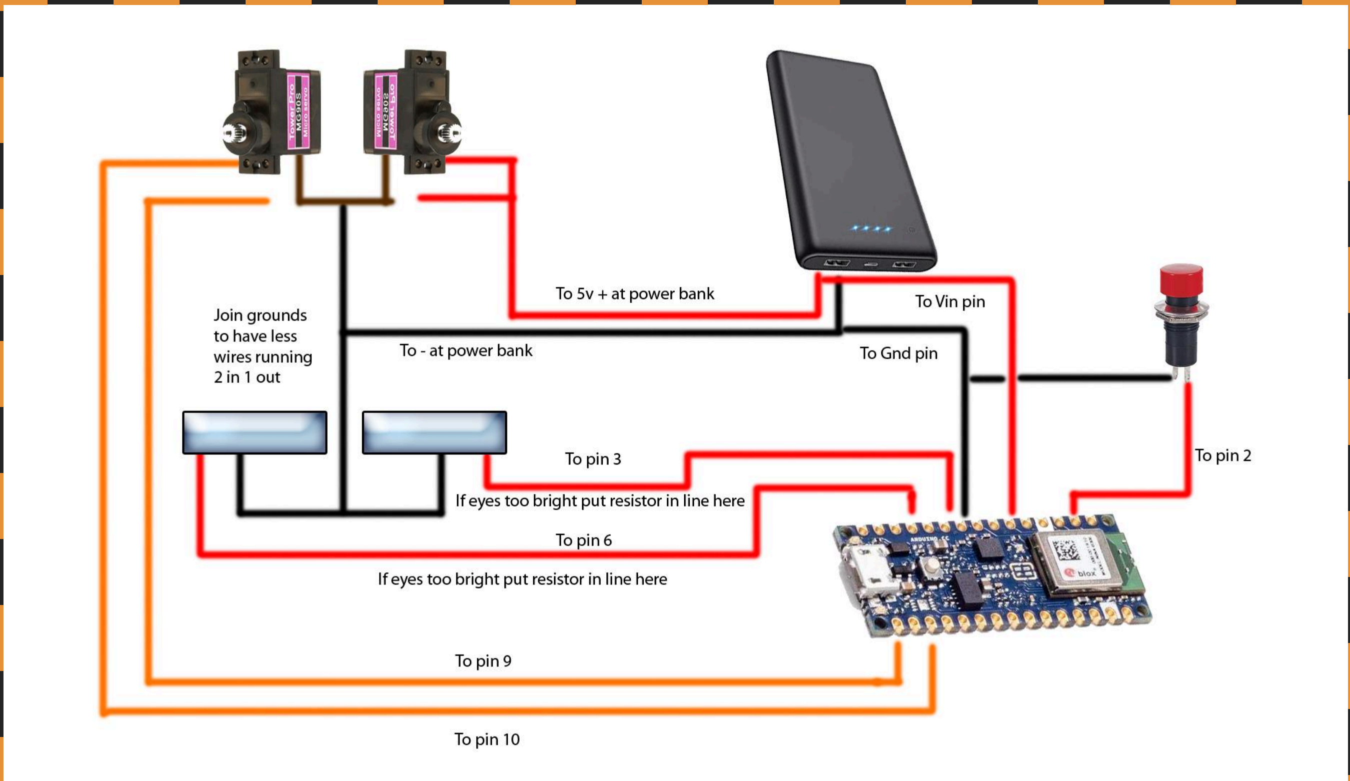

Via Crashworks3D Iron-Man Servo

For the code, "A.L.I.S.H.A. board, or other resources, it can be found on Crashworks "Iron-Man Servo" GitHub Page

Electronics Resources Link	
Website:	https://www.crashworks3d.com/
GitHub:	https://github.com/crashworks3d/Iron_Man_Servo?tab=readme-ov-file

Mounting to Other Helmets

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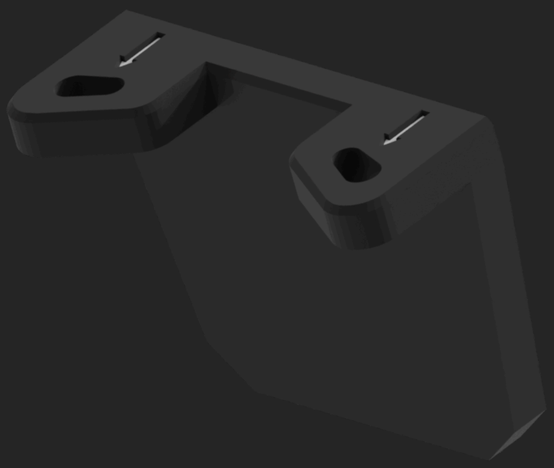
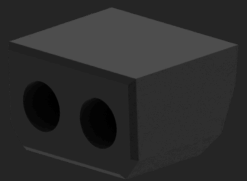
This section will cover adding the mount to non-compatible helmets. There are instructions for adding the mount in blender, and adding the mount post-printing.



In the folder, there are several files titled "Mounting System." Shown on the right.

The dome mount and faceplate mount *must* stay this exact distance apart so that the kit can be mounted.

It is highly recommended to use the Blender attachment method.



Blender Instructions

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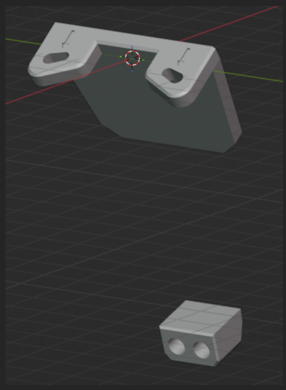
First, let's learn some basic blender controls:

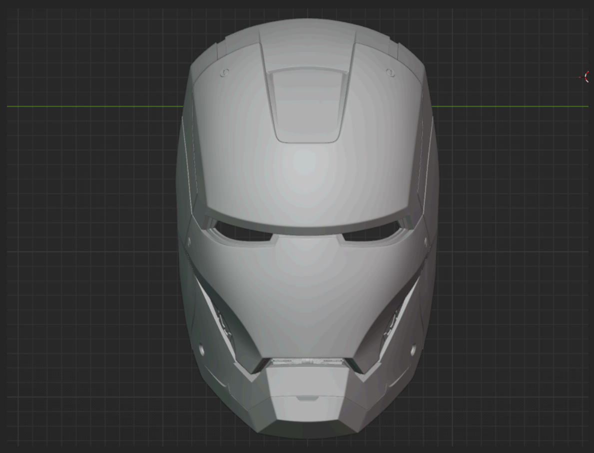
Blender Controls	
Keybind	**Action**
Middle Click	Rotate View
Shift + Middle Click	Pan
Tab	Switch between "Edit Mode" and "Object Mode"
G + X Y or Z	Move object in X Y or Z axis
R + X Y or Z	Rotate object in X Y or Z axis
E	Extrude

Now, let's jump right in.

Blender Mounting Instructions		
Step	**Description**	**Pictures**
1]	Open the "Mounting System" .blend file and become familiar with your <u>Do not move the mount from this position</u>. The two mounts need to stay the exact same distance from each other	

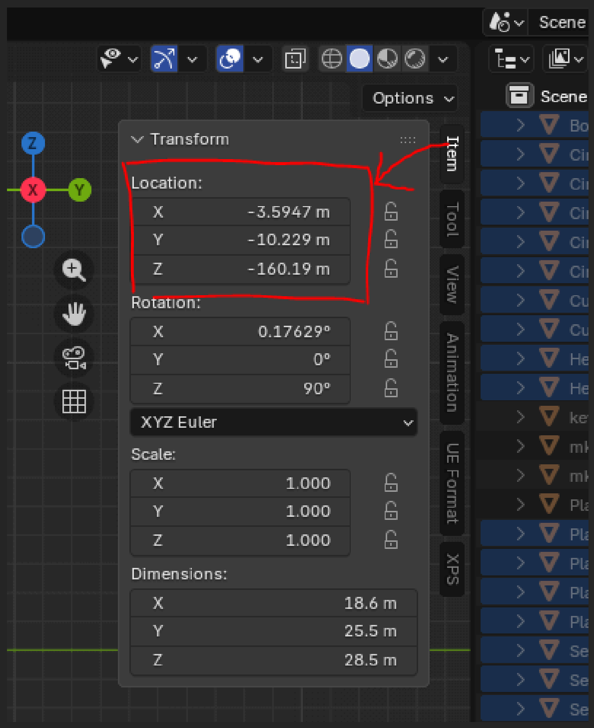
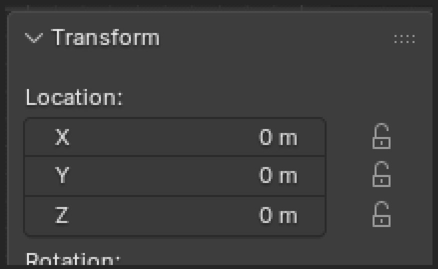
<table>
<tr>
<td>2]</td>
<td>In the top left, go to:

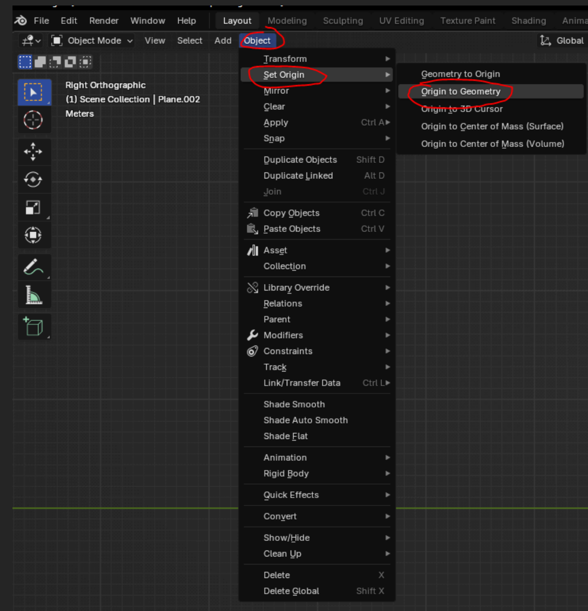
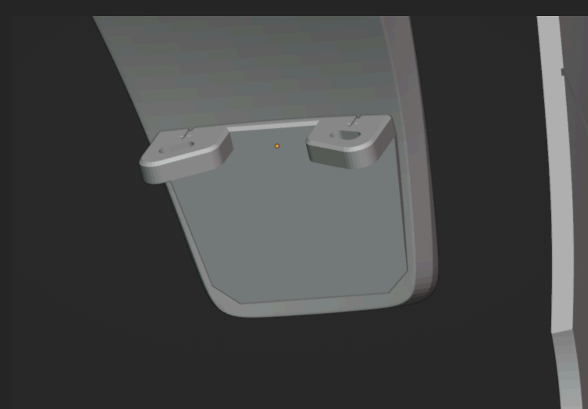
File -> Import -> STL and import your helmet

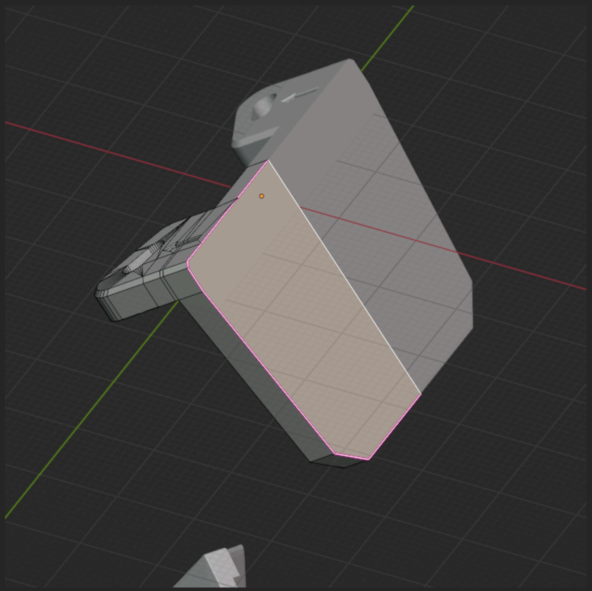
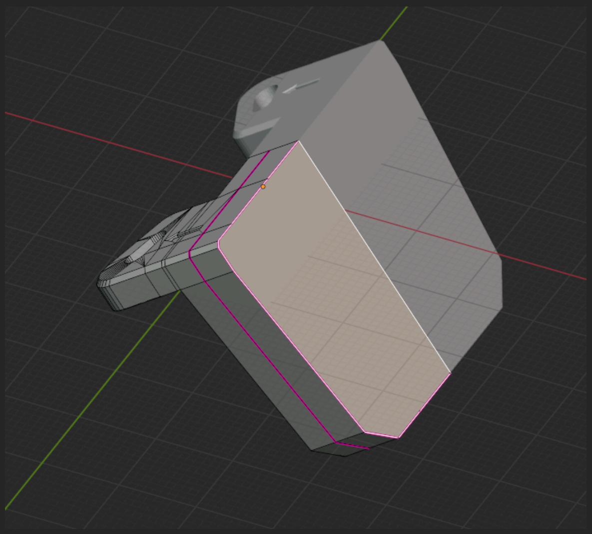
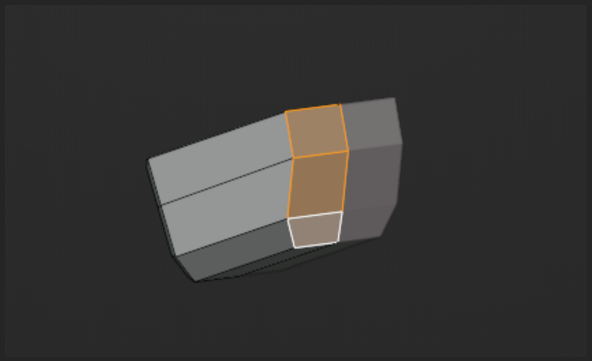
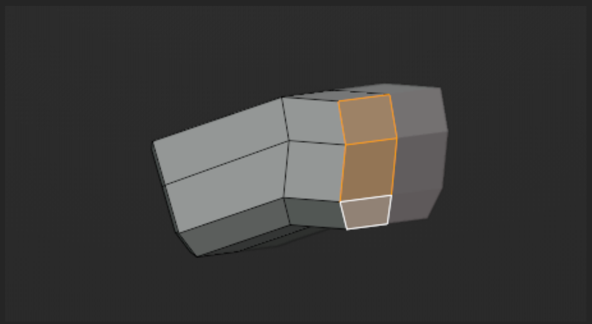
Arrange the pieces of the helmet so that the faceplate is in the correct place.</td>
<td></td>
</tr>
<tr>
<td>3]</td>
<td>Sometimes when you import an STL file into Blender, it shows up really far away from the center of the scene because of the position info saved in the file.

If you can’t see your helmet after it’s imported, you can select your helmet and set its X, Y, and Z position to 0.

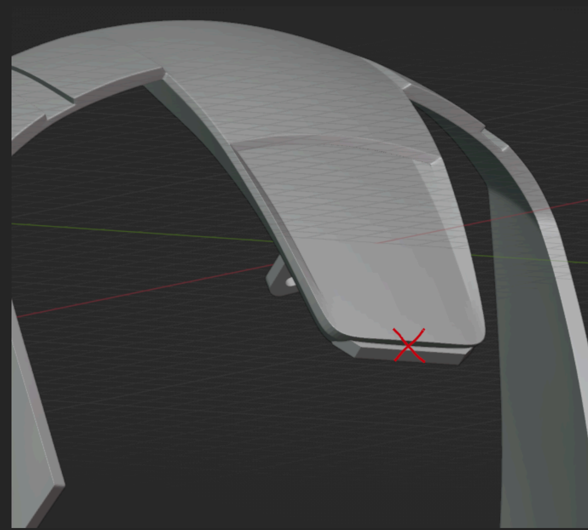
This menu is on the right in “Item” -> “Transform” -> “Location”

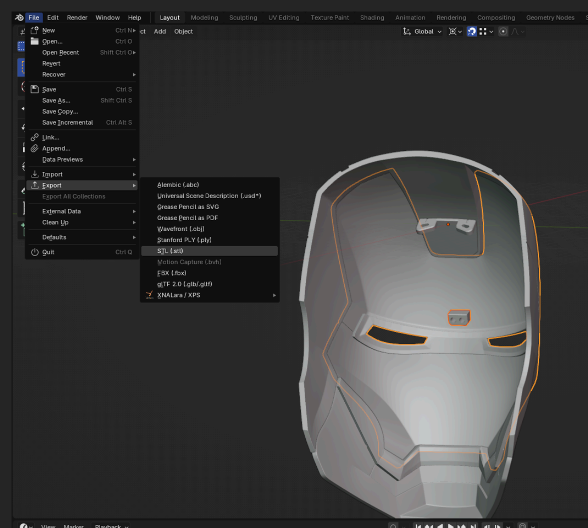
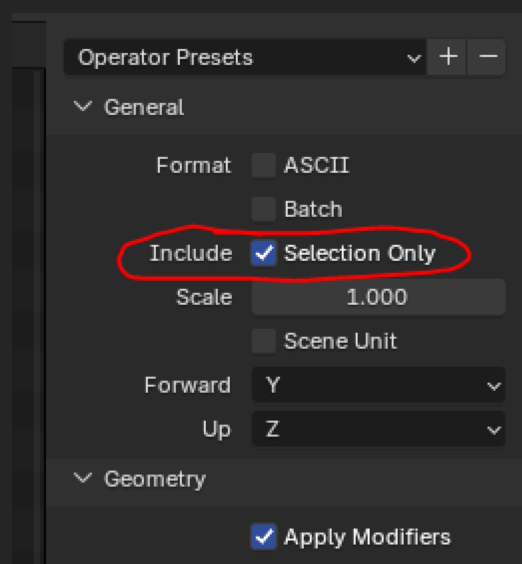
It should read:
X 0 m
Y 0 m
Z 0 m</td>
<td>
</td>
</tr>
<tr>
<td>4]</td>
<td>It should look like this:</td>
<td>
</td>
</tr>
</table>

5]	If that didn’t work, While the object is still selected, navigate to “Object” in the top left and do: “Object” -> “Set Origin” -> “Origin to Geometry” Now, repeat step 3 and 4 for any object you cannot see	
6]	Use your move (“G”) and rotate (“R”) tools to position your helmet so that the dome and faceplate intersect the mounting system Try your best to center the helmet over the mount Use X Y and Z axis keys for rotating the helmet, that way it will be easier to control your movements	
7]	After positioning the helmet over the mount, make sure the bottom face of the mount isn’t intersecting with the helmet and is still completely flat	

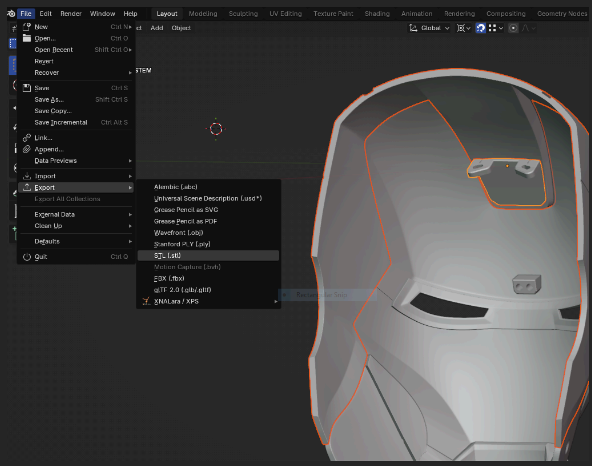
8]	If the Mounting System does not reach your dome, hide the dome and follow these steps: Click the mount and go into “Edit Mode” (“Tab”) Select the back face of it, and extrude (“E”)	
9]	This is how extruding will look Extrude as much as you need it without it coming through the other side of the helmet Now, return to “Object Mode” (“Tab”) <u>For most helmets, this is unnecessary</u>	
10]	If the Mounting System does not reach your faceplate, hide the faceplate and follow these steps: Click the mount and go into “Edit Mode” (“Tab”) Select the back 3 faces of it, and extrude (“E”)	
11]	This is how extruding will look Extrude as much as you need it without it coming through the other side of the faceplate Now, return to “Object Mode” (“Tab”) <u>For most helmets, this is unnecessary</u>	

<table>
<tr><td>12]</td><td>Now that your mount is intersecting with the helmet, verify that the dome mount is <u>NOT</u> poking through the edge of the widows peak.

This could potentially cause the faceplate not to fully close.</td><td></td></tr>
<tr><td>13]</td><td>Select JUST the faceplate mount and faceplate and do:

File -> Export -> STL (.stl)</td><td></td></tr>
<tr><td>14]</td><td>On the right of the pop-up window, select “Selection Only” and export</td><td>
</td></tr>
</table>

<table>
<tr>
<td>15]</td>
<td>Select JUST the dome mount and dome and do:

File -> Export -> STL (.stl)</td>
<td>
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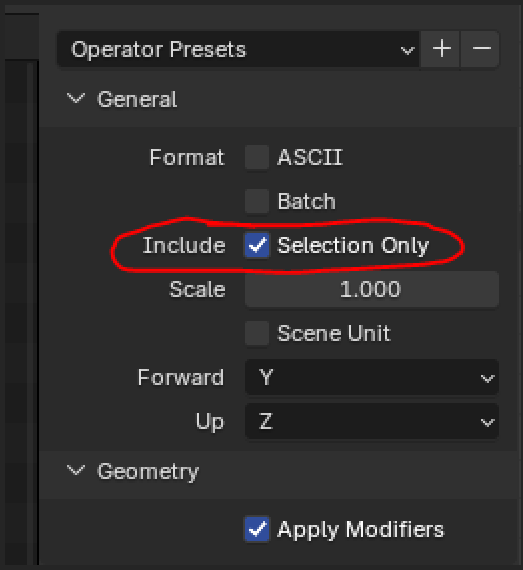
<td>16]</td>
<td>On the right of the pop-up window, select “Selection Only” and export</td>
<td>
</td>
</tr>
<tr>
<td>Done!</td>
<td>You’ve successfully added the mount to your helmet using Blender</td>
<td></td>
</tr>
</table>

Post-Printing Instructions

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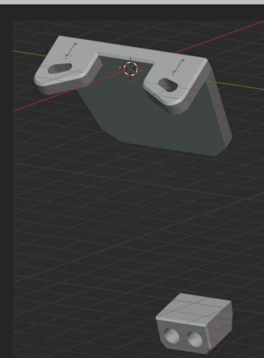
Unfortunately, to attach the mount after you already have a printed helmet, you still need to use a little bit of Blender to make sure the mount perfectly fits.

So first, let's learn some basic blender controls:

Blender Controls	
Keybind	**Action**
Middle Click	Rotate View
Shift + Middle Click	Pan
Tab	Switch between "Edit Mode" and "Object Mode"
G + X Y or Z	Move object in X Y or Z axis
R + X Y or Z	Rotate object in X Y or Z axis
E	Extrude

Now, let's jump right in.

Blender Mounting Instructions		
Step	**Description**	**Pictures**
1]	Open the "Mounting System" .blend file and become familiar with your <u>Do not move the mount from this position</u>. The two mounts need to stay the exact same distance from each other	

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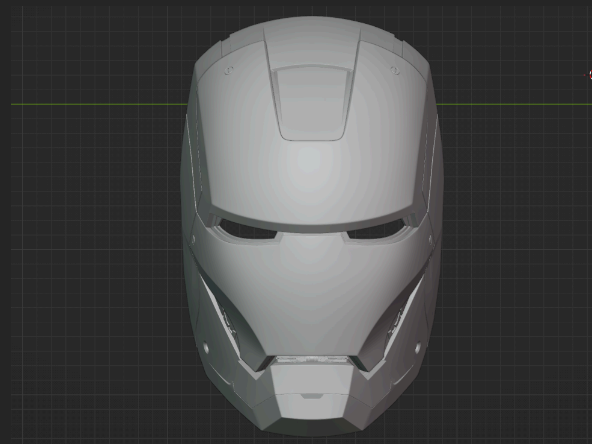
<td></td>
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<td>3]</td>
<td>Sometimes when you import an STL file into Blender, it shows up really far away from the center of the scene because of the position info saved in the file.

If you can’t see your helmet after it’s imported, you can select your helmet and set its X, Y, and Z position to 0.

This menu is on the right in “Item” -> “Transform” -> “Location”

It should read:
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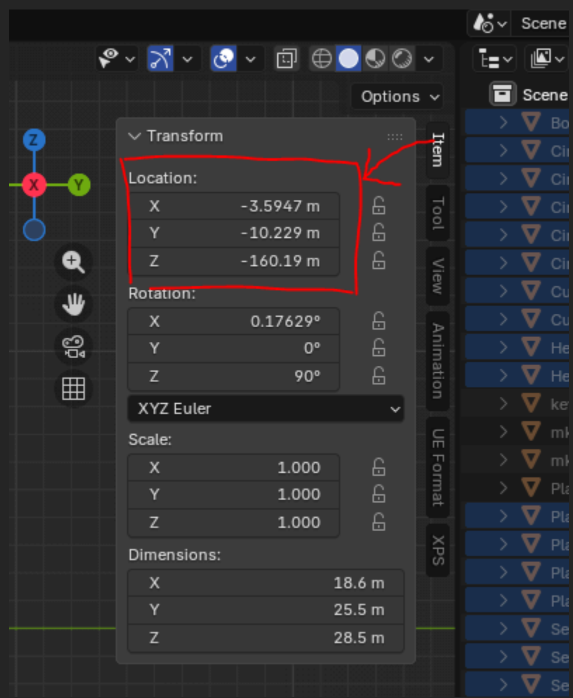
Y 0 m
Z 0 m</td>
<td>
</td>
</tr>
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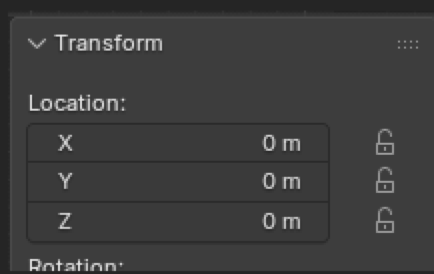
<td>4]</td>
<td>It should look like this:</td>
<td>
</td>
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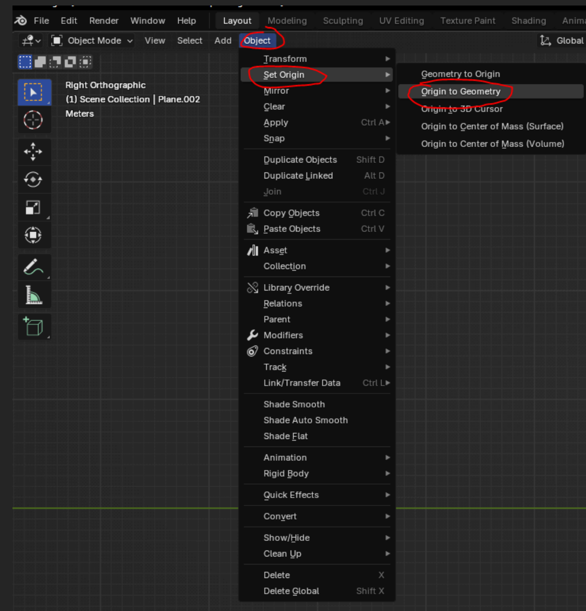
<table>
<tr>
<td>5]</td>
<td>If that didn't work,

While the object is still selected, navigate to "Object" in the top left and do:

"Object" -> "Set Origin" -> "Origin to Geometry"

Now, repeat step 3 and 4 for any object you cannot see</td>
<td>
</td>
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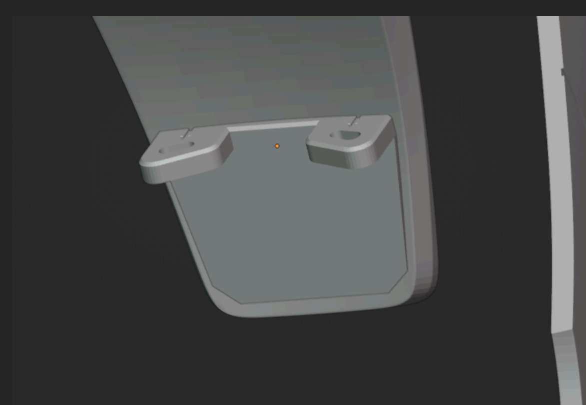
<td>6]</td>
<td>Use your move ("G") and rotate ("R") tools to position your helmet so that the dome and faceplate intersect the mounting system

Try your best to center the helmet over the mount

Use X Y and Z axis keys for rotating the helmet, that way it will be easier to control your movements</td>
<td></td>
</tr>
<tr>
<td>7]</td>
<td>After positioning the helmet over the mount, make sure the bottom face of the mount isn't intersecting with the helmet and is still completely flat</td>
<td></td>
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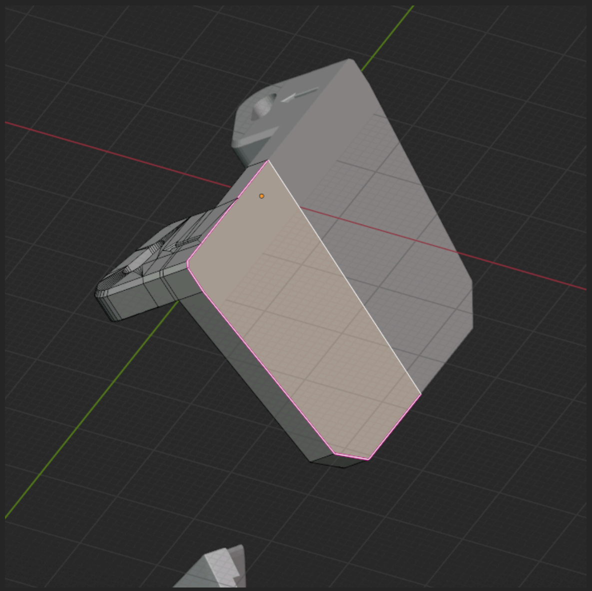
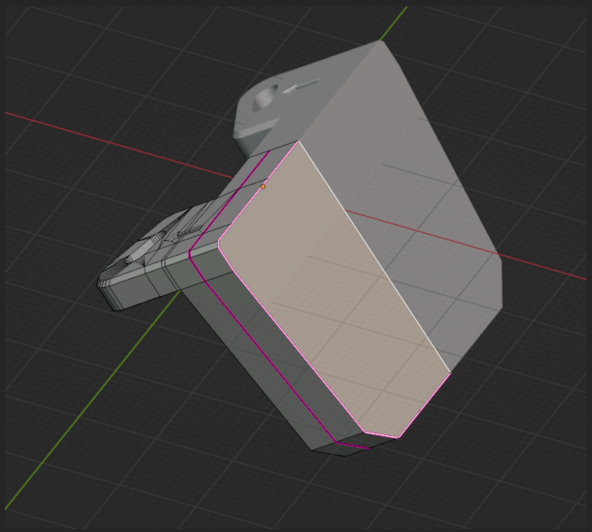
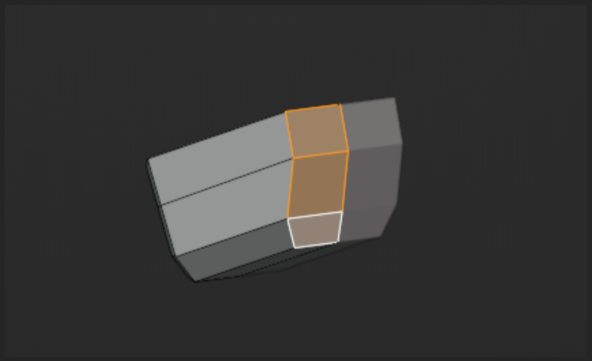
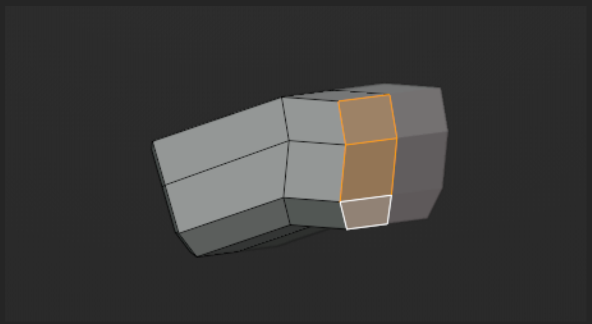
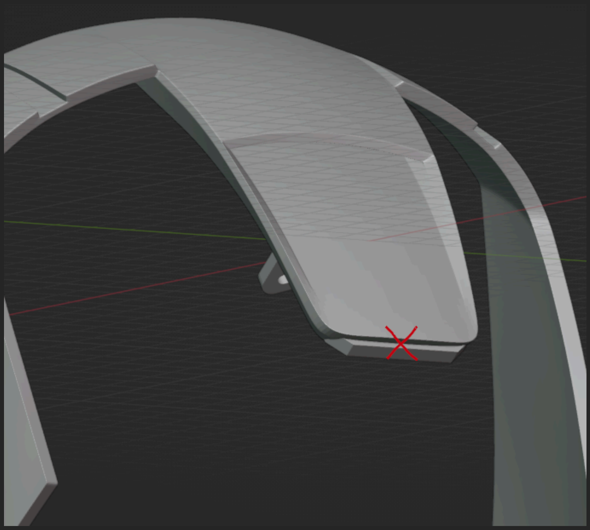
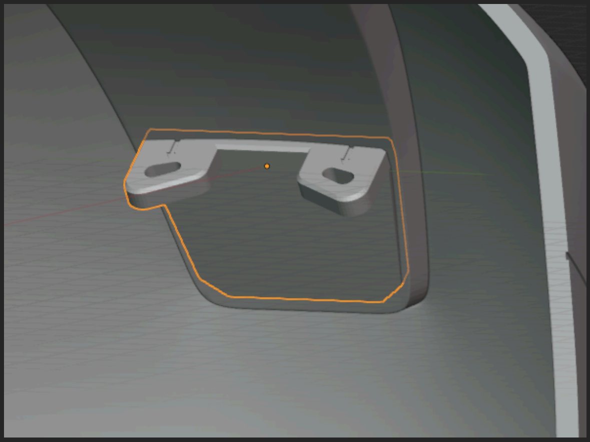
</table>

8]	If the Mounting System does not reach your dome, hide the dome and follow these steps: Click the mount and go into “Edit Mode” (“Tab”) Select the back face of it, and extrude (“E”)	
9]	This is how extruding will look Extrude as much as you need it without it coming through the other side of the helmet Now, return to “Object Mode” (“Tab”) <u>For most helmets, this is unnecessary</u>	
10]	If the Mounting System does not reach your faceplate, hide the faceplate and follow these steps: Click the mount and go into “Edit Mode” (“Tab”) Select the back 3 faces of it, and extrude (“E”)	
11]	This is how extruding will look Extrude as much as you need it without it coming through the other side of the faceplate Now, return to “Object Mode” (“Tab”) <u>For most helmets, this is unnecessary</u>	

<table>
<tr>
<td>12]</td>
<td>Now that your mount is intersecting with the helmet, verify that the dome mount is <u>NOT</u> poking through the edge of the widows peak.

This could potentially cause the faceplate not to fully close.</td>
<td></td>
</tr>
<tr>
<td>13]</td>
<td>Now, let's fit the dome mount perfectly to your helmet.

First, select the dome mount</td>
<td></td>
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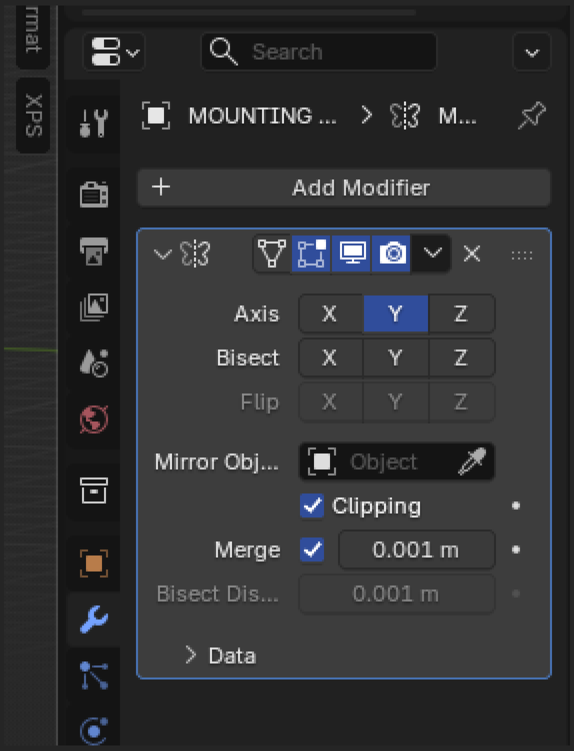
<td>14]</td>
<td>Navigate to the menu on the right, and select the blue wrench icon.

Then, select "+ Add Modifier" and find "Boolean"</td>
<td>
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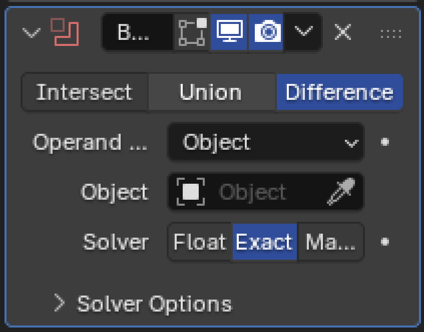
<td>15]</td>
<td>In your boolean modifier, use the “Object” eye-dropper tool and select the dome of your helmet.

This will perfectly cut the dome mount to match the shape of your helmet’s dome.</td>
<td>
</td>
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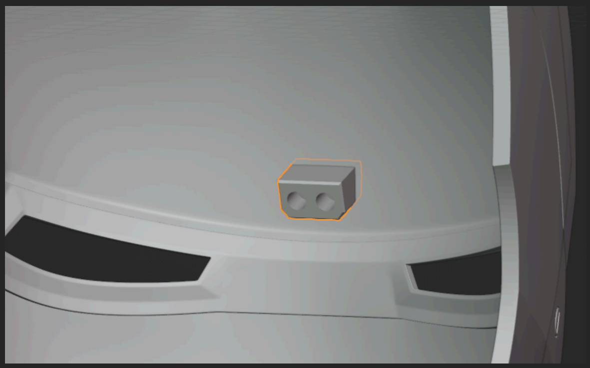
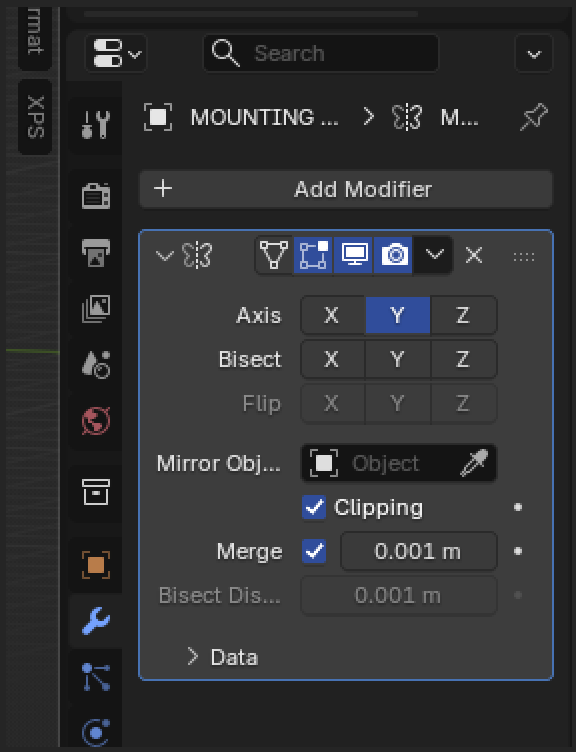
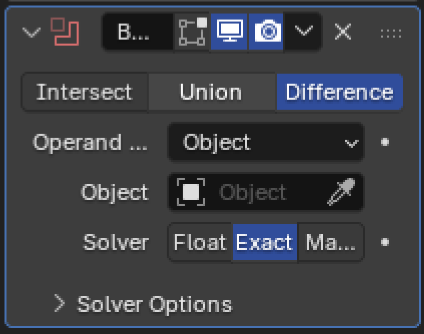
<td>16]</td>
<td>Now, let’s fit the faceplate mount perfectly to your faceplate.

First, select the faceplate mount</td>
<td></td>
</tr>
<tr>
<td>17]</td>
<td>Navigate to the menu on the right, and select the blue wrench icon.

Then, select “+ Add Modifier” and find “Boolean”</td>
<td>
</td>
</tr>
<tr>
<td>18]</td>
<td>In your boolean modifier, use the “Object” eye-dropper tool and select the faceplate of your helmet.

This will perfectly cut the faceplate mount to match the shape of your helmet’s faceplate.</td>
<td>
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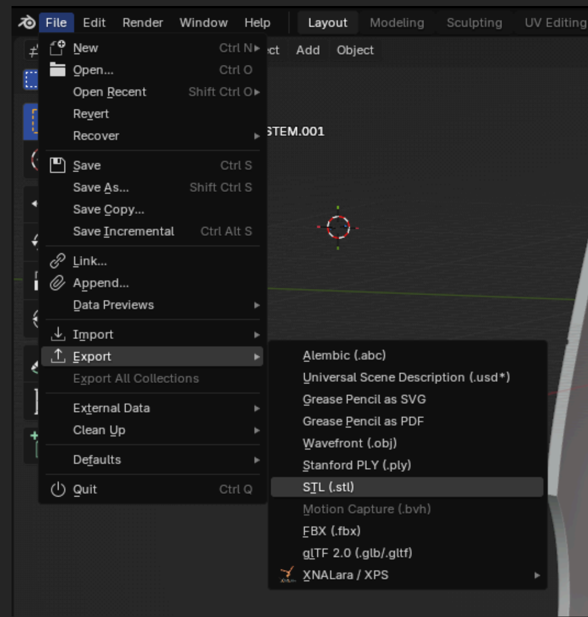
<td>19]</td>
<td>Select JUST the faceplate mount and do:

File -> Export -> STL (.stl)</td>
<td>
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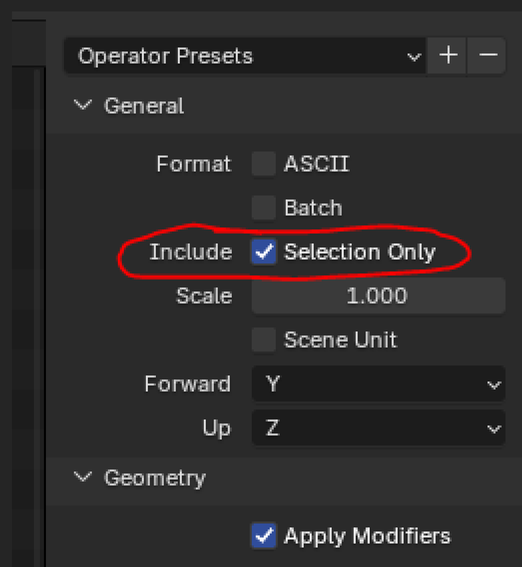
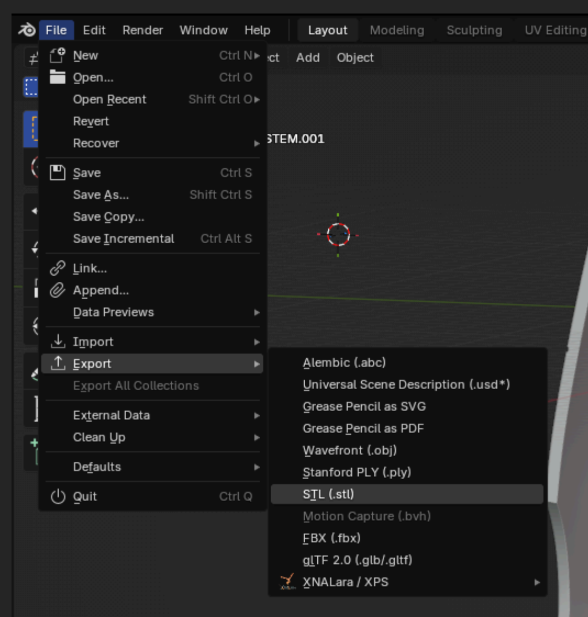
<td>20]</td>
<td>On the right of the pop-up window, select “Selection Only” and export</td>
<td>
</td>
</tr>
<tr>
<td>21]</td>
<td>Select JUST the dome mount and do:

File -> Export -> STL (.stl)</td>
<td>
</td>
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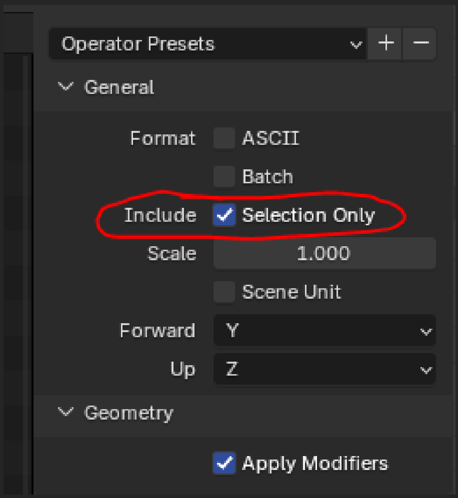
</table>

22]	On the right of the pop-up window, select “Selection Only” and export	
23]	Now, print those two pieces, along with the provided “alignment tool”	
24]	Use the “alignment tool” to align your two parts like this. Apply glue to the back of both “Mounting System” pieces, and stick them into your helmet.	
Done!	Now it is mounted in your helmet and you can remove the alignment tool.	

Usage Permission

"Motorization"
VER - NO. 01

Any 3D designer is free to design helmets or accessories intended to work with this motorization kit.

The "mounting system" may be included directly in your own 3D models and distributed with them.

However, the motorization kit itself must not be rehosted or redistributed. If your design relies on this kit, you must include a clear link to the original listing where the motorization files can be downloaded.

> Attribution is appreciated but not required beyond the linking requirement above.

Best,
Walsh3D